



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI

INHA UNIVERSITY IN TASHKENT

–1 POINTS FOR EACH FIELD LEFT EMPTY OR FILLED WRONG

ID NUMBER: NAME:
SECTION: ROOM NUMBER

FINAL EXAMINATION

COURSE NAME: COURSE NUMBER:

EXAMINATION DURATION:

ADDITIONAL MATERIALS
ALLOWED TO BE USED:

SPECIAL INSTRUCTIONS

(1) FOLLOW ANY INSTRUCTIONS GIVEN BY PROCTORS.
(2) ONCE YOU ARE ALLOWED TO OPEN THE EXAMINATION BOOKLET,
YOU MUST PUT YOUR **ID** ON EVERY ODD PAGE
(3) READ THE INSTRUCTION CAREFULLY!
Sometimes it is asked to provide JUST an answer, sometimes **FULL** solution.

- Final answers must be written by **only blue, non-erasable pen**.
- Do NOT use highlighters or correction pen.
- IUT's Examination Rules and Regulations apply.

Do NOT open the examination paper until directed to do so!

<u>Do NOT fill</u>	<u>P.1</u>	<u>P.2</u>	<u>P.3</u>	<u>P.4</u>	<u>P.5</u>	<u>P.6</u>	Total
max points	10	8	7	6	4	5	40
your score							

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1. (10 points) If $a_n = 10a_{n-1} - 25a_{n-2} + 5^n$ is a recurrence relation with initial conditions $a_0 = 1, a_1 = 12.5$ solve it following the steps.

a) (3 points) find all solutions of homogeneous part.

b) (4 points) find a particular solution of non-homogeneous part.

- c) (3 points) combine homogeneous and non-homogeneous solutions and determine unknown constants.

2. (8 points) Which of these collections of subsets are partitions of $\{1, 2, 3, 4, 5, 6\}$?

- a) $\{1, 2\}, \{2, 3, 4\}, \{4, 5, 6\}$
- b) $\{1\}, \{2, 3, 6\}, \{4\}, \{5\}$
- c) $\{1, 3, 5\}, \{2, 4, 6\}$
- d) $\{1, 4, 5\}, \{2, 6\}$

Justify your answers below by providing corresponding relation (pairs).

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3. (7 points) Schedule the final exams for Math 115, Math 116, Math 185, Math 195, CS 101, CS 102, CS 273, and CS 473, using the fewest number of different time slots, if there are no students taking both Math 115 and CS 473, both Math 116 and CS 473, both Math 195 and CS 101, both Math 195 and CS 102, both Math 115 and Math 116, both Math 115 and Math 185, and both Math 185 and Math 195, but there are students in every other pair of courses. Apply graph coloring.

4. (4 points) Connect the sentences halves.

- | | |
|---|---------------|
| • A tree with 26 vertices has | • 143 leaves |
| • A full binary tree with 13 leaves vertices has | • 25 edges |
| • If a full 5-ary tree has 10 edges then it has | • 25 vertices |
| • A full 3-ary tree with 71 internal vertices has | • 9 leaves |

5. (6 points) Let the postfix expression $32 * 2 \uparrow 53 - 84 / * -$ be given where $\uparrow$ means taking a power. Each number in the expression is one-digit number (strictly less than 10).

a) (4 points) provide its tree.

b) (2 points) find its value.

6. (5 points) Draw minimum spanning tree of the graph and find its weight.

